KAMALESH SURESH KUMAR

FULL-STACK SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Computer Science undergraduate at Rajalakshmi Engineering College (Expected 2027) and Full-Stack Software Engineer with proven ability in architecting decoupled, high-performance web applications, modern component architectures (React 19, TypeScript), and automated deployment pipelines. Creator of KAI HQ, a fully client-side portfolio operating system featuring deterministic state management, sub-10ms search capabilities, and 100% static hosting on GitHub Pages.

TECHNICAL SKILLS & CORE COMPETENCIES

Languages: TypeScript, JavaScript (ESNext), Python, HTML5, CSS3, SQL
Frameworks: React 19, Node.js & Express, Tailwind CSS, Vite
Tools & DevOps: Git, GitHub Actions (CI/CD), Linux/Unix, Bash Scripting
Key Competencies: Full-Stack Web Architecture * Component Design & Custom Hooks * State Management & Mediators * RESTful API & Service Integration * Performance Optimization * WCAG 2.1 AA Accessibility

KEY ENGINEERING PROJECTS

KAI HQ — Headquarter Operating System | *React 19, TypeScript, Tailwind CSS, Vite, GitHub Pages*
- Architected an 8-room interactive portfolio operating system replacing generic portfolio patterns with a decoupled Experience and Knowledge Engine architecture.
- Implemented clean client-side path routing, popstate navigation, and 404 fallback restoration for production deployment on GitHub Pages.
- Engineered a persistent shell with fixed header and sidebar, dedicated scrolling container, and accessible mobile navigation drawer.

Automated CI/CD Knowledge & Build Pipeline | *GitHub Actions, Node.js, Vite, Bash, YAML*
- Developed automated continuous integration and continuous deployment workflows running automated typechecking, linting gates, and static asset publication.
- Implemented multi-environment path resolution enabling dynamic runtime adaptation between local development and GitHub Pages subpaths.

Rule-Based Conversational Router & Knowledge Dispatcher | *TypeScript, Natural Language Intent Resolution, Algorithms*
- Designed a deterministic, client-side natural language intent parser routing visitor queries to spatial coordinates and entities in under 10ms with zero API latency.

EXPERIENCE & LEADERSHIP

Software Development Intern — Viyuga Innovations *(2024 | India)*
- Engineered reusable modular components and responsive user interface flows following enterprise engineering standards.
- Participated in active code reviews, sprint workflows, and frontend performance profiling.

Virtual Intern / Technical Trainee — Infosys SpringBoard *(2023 — 2024 | Virtual / India)*
- Completed structured virtual internship training covering cloud computing principles, enterprise software engineering methodologies, and structured algorithmic problem solving.

Naval Wing Cadet & Field Leadership — National Cadet Corps (NCC - Navy Wing) *(2022 — Present | India)*
- Underwent operational discipline, naval procedures, drill precision, and crisis response teamwork.

EDUCATION

B.E. in Computer Science and Engineering — Rajalakshmi Engineering College *(2022 — 2027 (Expected) | India)*
- Core Coursework: Systems Architecture, Data Structures, Algorithms, Operating Systems, Computer Networks, Database Systems, Network Security.

CERTIFICATIONS & VERIFIED HONORS

- Enterprise Software & Cloud Computing Certification — Infosys SpringBoard (2023 — 2024)
- Li-Fi Penetration Testing Demonstration & Law Enforcement Briefing — Cybercrime Police Department (2024)
- NCC Naval Wing Cadet Leadership & Service — National Cadet Corps (2022 — Present)
- IEEE Computer Society Event Team Contribution Award — IEEE Student Chapter (2023 — 2024)